Customer Shopping Behavior Analysis

Business Problem Statement & Analytical Scope

Project workflow: Jupyter Notebook (Python) → Microsoft SQL Server, managed with SQL Server Management Studio (SSMS) 22 → Power BI Desktop.

Project context

A retail company wants to better understand customer shopping behavior to support decisions about sales, customer satisfaction, and long-term loyalty. The problem focuses on differences in purchasing patterns across customer demographics and product categories, and asks whether factors such as discounts, reviews, seasons, and payment preferences relate to consumer decisions and repeat purchases.

Source scope note: The supplied CSV contains 3,900 rows and 18 columns. It includes customer demographics, product/category, purchase amount, season, discount and promotion flags, previous purchases, purchase-frequency category, review rating, subscription status, and shipping type. The supplied fields do not include a purchase date, an explicit online/offline channel, or a payment-preference field. Therefore, those topics cannot be directly tested using this file without additional data.

1. Core Business Question

How can the company use the available customer-shopping records to describe purchasing patterns and identify useful opportunities to review product merchandising, discount practices, subscription engagement, shipping options, and customer-retention efforts?

2. Business Objectives

- Summarize purchase amounts and record counts across available customer groups.
- Compare descriptive purchase metrics by gender, age group, subscription status, and selected shipping types.
- Identify products with high average available ratings, high purchase counts, or high discount frequency.
- Explore how previous-purchase counts relate to the project's customer-segment labels and subscription status.
- Present results in a reproducible workflow using Python, SQL Server, SSMS 22, and Power BI Desktop.

3. Analytical Questions

No.	Analytical question	Primary fields / approach
1	How does total purchase amount differ by gender?	gender, purchase_amount; group totals and averages
2	Which discounted records have purchase amounts at or above the overall average?	discount_applied, purchase_amount; filter against dataset average
3	Which products have the highest average available review ratings?	item_purchased, review_rating; average excluding null ratings
4	How do Standard and Express shipping records compare in average purchase amount?	shipping_type, purchase_amount; selected shipping types only
5	How do purchase count, average amount, and total amount differ by subscription status?	subscription_status, purchase_amount
6	Which products have the highest share of records marked discounted?	item_purchased, discount_applied; percentage within product
7	How are records distributed under the supplied New / Returning / Loyal rule?	previous_purchases; SQL CASE rule

No.	Analytical question	Primary fields / approach
8	Which three items have the most records in each product category?	category, item_purchased; rank counts within category
9	How many records with more than five previous purchases are subscribers and non-subscribers?	previous_purchases, subscription_status
10	How do purchase amounts vary across age groups?	age_group, purchase_amount; grouped summary

4. Data and Method

The workflow starts from *customer_shopping_behavior.csv*. The Python notebook in Jupyter Notebook inspects and standardizes columns, creates an age_group feature and an approximate purchase_frequency_days mapping, checks whether discount_applied and promo_code_used match, and loads the transformed data into a Microsoft SQL Server table. SQL analysis is run using the supplied SQL script in SQL Server Management Studio (SSMS) 22. The script contains the ten analytical queries listed above. Power BI Desktop is the intended visual presentation layer; its figures should be checked against executed SQL outputs before publication.

5. Scope and Assumptions

- The analysis is descriptive and based on the fields in the supplied CSV; group differences should not be described ascausal effects.
- Purchase amount is used as a monetary transaction measure. Profit, margin, units sold, customer lifetime value, andincremental campaign impact are outside the available data.
- The notebook's purchase-frequency mapping is an approximate mapping from labels, not a calculation from datedtransaction history.
- The supplied SQL's customer-segment definitions are retained for reproducibility, but should be reviewed withstakeholders before being used operationally.
- Review-rating averages should be calculated from available non-null ratings unless an explicit imputation step is addedand rerun.

6. Out of Scope / Additional Data Needed

Topic in original brief	Current support in supplied data	What would be needed
Online vs offline channel	No explicit channel field identified	Channel indicator and consistent definitions
Payment preferences	No payment method field identified	Payment method/category in transaction records
Seasonal purchase changes over time	Season field exists, but no purchase date	Purchase dates and multiple periods for trend analysis
Discount impact on sales/profit	Discount flag and purchase amount exist	Control/comparison design, price, cost/margin, campaign exposure
Repeat-purchase timing	Previous-purchase count and frequency label exist	Customer-level dated purchase history

7. Deliverables

1. Python notebook for data inspection, preparation, and SQL Server loading (run in Jupyter Notebook).
2. SQL script with the ten business-analysis queries (execute in Microsoft SQL Server using SSMS 22).
3. Power BI Desktop report with validated visuals and filters.
4. Analysis report and presentation summarizing supported findings and recommendations.
5. Well-structured GitHub repository containing the project artifacts and usage notes.

8. Success Criteria

- All reported metrics can be reproduced from the supplied CSV and executed SQL queries.

- Data-quality decisions, missing-value handling, feature definitions, and segment rules are documented.
- Dashboard totals reconcile with the source dataset and SQL outputs.
- Conclusions distinguish observed patterns from hypotheses that require additional data or testing.
- Repository instructions allow another person to understand the files and reproduce the workflow.